Athena++ Simulation Campaign Specifications
Filament Fragmentation Study

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1 Overview

This document provides complete Athena++ problem generator specifications for all filament fragmentation campaigns reported in the associated paper. These specifications enable exact reproduction of the simulations.

Total campaigns: 8 distinct campaigns totaling 2,000+ simulations

- Field Geometry Campaign (314 simulations)
- Definitive Transition Campaign (540 simulations)
- Supercritical Filament Campaign (654 simulations)
- Critical Transition Zone Mapping (CTZM, 96 simulations)
- Turbulence and Field Geometry (289 simulations)
- Validation Campaigns (83 simulations)
- Perpendicular-Field Campaign (96 simulations)
- Regime Transition Investigations (additional)

2 Common Parameters

2.1 Units and Normalization

All simulations use Athena++ normalised units:

- Gravitational constant: `four_pi_G` $= 4\pi^2$ (sets $\lambda_J = 1$)
- Sound speed: $c_s = 1.0$
- Characteristic Jeans wavelength: $\lambda_J = 1$
- Core width: $W_{\mathrm{core}} = 0.3\,\lambda_J$
- Time unit: $t_{\mathrm{ff}} = 1/\sqrt{G\rho_0}$

2.2 Default Boundary Conditions

```
<boundary>
    bc_x1 = periodic     # periodic longitudinal
    bc_x2 = outflow       # outflow transverse y
    bc_x3 = outflow       # outflow transverse z
</boundary>
```

2.3 Default Mesh Configuration

Unless otherwise specified:

- Resolution: 256^3 cells ($128 \times 64 \times 64$ minimum)
- Domain: $8\lambda_J \times 2\lambda_J \times 2\lambda_J$
- Mesh refinement: AMR disabled
- Coordinate system: Cartesian (x_1, x_2, x_3)

3 Campaign 1: Field Geometry Campaign

3.1 Purpose

Test magnetic field geometry effects (longitudinal vs perpendicular vs oblique) on fragmentation wavelength.

3.2 Parameter Grid

Parameter	Values	Count	Notes
Line-mass f	1.5–3.0	10 values	Supercritical ra
Plasma β	0.3–5.0	8 values	Weak to strong I
Mach $\mathcal{M}$	0.5–3.0	4 values	Subsonic to tran
Field angle θ	$0°, 30°, 45°, 60°, 90°$	5 values	Full range
Random seeds	2 per point	–	Stochasticity te
Total		**314**	

Table 1: Campaign 1 parameter grid

3.3 Athena++ Configuration

```
<job>
  problem      = filament   # Filament problem generator
  tlim         = 2.0       # Stop time (t_J)
  dt           = 0.001
</job>

<mesh>
  nx1 = 256    # Longitudinal
  nx2 = 64
  nx3 = 64
  x1min = 0.0
  x1max = 8.0    # L_x = 8 lambda_J
  x2min = -1.0
```

```
  x2max = 1.0    # L_y = 2 lambda_J
  x3min = -1.0
  x3max = 1.0    # L_z = 2 lambda_J
</mesh>

<hydro>
  iso_sound_speed = 1.0
</hydro>

<field>
  b_initial = uniform
  bx = B0 * cos(theta)
  by = 0.0
  bz = B0 * sin(theta)
  # B0 scaled from beta
</field>

<problem>
  rho0 = 1.0
  W_core = 0.3
  f = [line-mass fraction]
  beta = [plasma beta]
  M = [mach number]
  theta = [field angle degrees]
  seed = [random seed]
</problem>

<output>
  output_dir = output/
  file_type = hdf5
  dt = 0.05    # Snapshot interval
</output>
```

3.4 Initial Conditions

Gaussian-profile filament:

$$\rho(r) = \frac{\rho_0}{1 + (r/W_{\text{core}})^2} \tag{1}$$

where $r = \sqrt{x_2^2 + x_3^2}$ is radial distance from axis.

3.5 Magnetic Field Strength

B0 computed from plasma β:

$$B_0 = \sqrt{\frac{8\pi P_{\text{therm}}}{\beta}} \tag{2}$$

with $P_{\text{therm}} = \rho_0 c_s^2 = \rho_0$.

4 Campaign 2: Definitive Transition Campaign (DTC)

4.1 Purpose

Map fragmentation timescale boundary between prompt and delayed fragmentation in parameter space.

Parameter	Values	Count	Notes
Line-mass f	1.4–2.2	9 values	Supercritical transition
Plasma β	0.3–1.3	6 values	Weak to strong B-field
Mach $\mathcal{M}$	1–5	5 values	Turbulence variation
Random seeds	2 per point	–	Stochasticity testing
Total		**540**	

Table 2: DTC parameter grid

4.2 Parameter Grid

4.3 Runtime Configuration

```
<job>
  tlim         = 10.0      # Extended timeout (6 hour wall
  dt           = 0.001
</job>

<output>
  dt = 0.05
</output>
```

4.4 Targeted Re-runs (April 2026)

15 STABLE points at $\beta = 0.3$, $\mathcal{M} = 1$ re-run with:

- Extended timeout: 6 hours wall-clock
- Same initial conditions as original
- Result: All 15 points fragmented ($t_{\text{frag}} \in [1.02, 1.43]\, t_{\text{J}}$)

5 Campaign 3: Supercritical Filament Campaign

5.1 Purpose

Direct measurement of fragmentation spacing in supercritical regime ($f \geq 1.5$).

5.2 Parameter Grid

Parameter	Fiducial	Near-Crit	Extension	Note
Line-mass f	1.5–3.0	1.1–1.3	–	Main grid
	10 values	3 values	–	Supercritica
Plasma β	0.3–5.0	–	3.0, 5.0	8 values
Mach $\mathcal{M}$	0.5–3.0	–	–	4 values
Random seeds	2 per point	–	–	–
Fiducial	126	–	–	$f = 1.5$–3.0
Near-crit	–	108	–	$f = 1.1$–1.3
High-β	–	–	84	$\beta = 3.0, 5.0$
Low-$\mathcal{M}$	–	–	84	$\mathcal{M} = 0.5$
Total	**654**	**108**	**168**	

Table 3: Supercritical campaign breakdown

5.3 Snapshot Strategy

Fiducial grid: Dense HDF5 output with $\Delta t = 0.05\,t_{\mathrm{J}}$

Analysis limitation: Transient beading at $f = 1.5$ develops over $\sim 0.03\,t_{\mathrm{J}}$, which is missed by 0.05 interval. CTZM campaign with final-snapshot analysis captures this.

6 Campaign 4: Critical Transition Zone Mapping (CTZM)

6.1 Purpose

Bridge gap between near-critical ($f \lesssim 1.2$) and supercritical ($f \gtrsim 1.5$) regimes with high-resolution sampling.

6.2 Parameter Grid

Parameter	Values	Count	Notes
Line-mass f	$1.2, 1.3, 1.4, 1.5$	4	Transition zone
Plasma β	$0.3, 0.5, 1.0, 2.0$	4	Weak to strong
Mach $\mathcal{M}$	$1.0, 2.0$	2	–
Random seeds	3 per point	6	High statistics
Total	**96**		

Table 4: CTZM parameter grid

6.3 Key Result

All 96 simulations (100%) exhibited longitudinal beading. No radial collapse observed in transition zone. λ/W evolves smoothly and monotonically with f, validating extrapolation.

7 Campaign 5: Perpendicular-Field Validation

7.1 Purpose

Independent validation of CTZM beading at $f = 1.5$ using perpendicular field geometry.

7.2 Parameter Grid

Parameter	Values	Count	Notes
Line-mass f	1.2–1.5	4	CTZM overlap
Plasma β	$0.3, 0.5, 1.0, 1.5, 2.0$	5	Full range
Random seeds	5 per point	–	High statistics
Field angle θ	$90°$	–	Perpendicular only
Total	**96**		

Table 5: Perpendicular-field campaign

7.3 Domain

Extended longitudinal domain for perpendicular-field analysis:

```
<mesh>
  nx1 = 512    # Extended for wavelength measurement
  nx2 = 64
  nx3 = 64
  x1min = 0.0
  x1max = 16.0   # L_x = 16 lambda_J
</mesh>
```

7.4 Key Result

At $f = 1.5, \beta = 2.0$: Measured $\lambda/W = 2.83 \pm 0.27$, matching CTZM longitudinal-field result ($\lambda/W = 2.86 \pm 0.29$) to within 1%.

8 Campaign 6: Turbulence Real vs Synthetic

8.1 Purpose

Test whether realistic ISM turbulence ($\delta v/c_s = 1.0$) modifies fragmentation wavelength compared to synthetic perturbations.

8.2 Parameter Grid

Parameter	Values	Count	Notes
Line-mass f	$1.0, 1.1, 1.2$	3	Near-criti
Plasma β	$0.5, 1.0, 2.0$	3	–
Turbulence type	Kolmogorov	–	Real vs synt
Turbulence amplitude	$\delta v/c_s = 1.0$ vs 10^{-4}	2	Physical vs
Random seeds	2 per point	–	–
Total	**36**		

Table 6: Turbulence comparison campaign

8.3 Initial Conditions

Synthetic perturbations:

```
<problem>
  turb_amplitude = 1e-4   # Weak seed
  turb_seed = [random seed]
</problem>
```

Physical turbulence:

```
<problem>
  turb_amplitude = 1.0    # ISM strength
  turb_spectrum = kolmogorov
  turb_mode = decaying
</problem>
```

8.4 Key Result

Difference in λ/W: $< 5\%$ (not statistically significant). Turbulence affects timescales but not wavelengths.

9 Campaign 7: EOS Sensitivity

9.1 Purpose

Test equation of state dependence on fragmentation wavelength across physically motivated range.

9.2 Parameter Grid

Parameter	Values	Count	Notes
Line-mass f	$1.0, 1.1, 1.2$	3	Near-critical
Plasma β	$0.5, 1.0, 2.0$	3	–
Adiabatic index γ	$0.7, 0.8, 0.9, 1.0$	4	Sub-isothermal to isothermal
Random seeds	3 per point	–	–
Total	**48**		

Table 7: EOS sensitivity campaign

9.3 EOS Configuration

Isothermal ($\gamma = 1.0$):

```
<hydro>
  iso_sound_speed = 1.0
</hydro>
```

Polytropic ($\gamma < 1$):

```
<hydro>
  eos = polytropic
  Gamma = [gamma value]
  T_floor = 0.01    # Prevent numerical issues
</hydro>
```

9.4 Key Result

λ/W varies by only 2.8% across γ = 0.7–1.0.$Nostatisticallysignificanttrend$(p=0.57).$Validatesisothermalapproximationforspacingmeasurements$.

10 Campaign 8: Resolution Convergence

10.1 Purpose

Verify that 128^3 resolution is adequate for fragmentation classification.

10.2 Test Matrix

10.3 Key Result

Mean ratio: $t_{\mathrm{frag}}(256^3)/t_{\mathrm{frag}}(128^3) = 0.928 \pm 0.016$ (7.2% difference, maximum 9.0%). Both resolutions show identical classification (all FRAG).

Resolution	f	β	$\mathcal{M}$	Seeds
128^3	$1.5, 2.0, 3.0$	1.0	–	Matched pairs
256^3	$1.5, 2.0, 3.0$	1.0	–	Matched pairs
Total	**6 pairs**	**12 simulations**		

Table 8: Resolution test matrix

11 Simulation Output Format

11.1 HDF5 Structure

```
output.hdf5
 times                    # Snapshot times (t_J)
 density                  # 3D density field (Nx, Ny, Nz)
 B_x, B_y, B_z            # Magnetic field components
 velocity_x, etc.         # Velocity components
 metadata
```

11.2 Peak Detection Method

Longitudinal beading detected using:

- Extract on-axis density profile: $\rho(x) = \langle \rho(x, y = 0, z = 0) \rangle$
- Find peaks with 'scipy.signal.find$_peaks$'$Prominencethreshold:$ $20\%ofpeakdensity$
- Minimum spacing: 10 grid cells

11.3 Fragmentation Wavelength Calculation

$$\lambda/W_{\mathrm{core}} = \frac{\Delta x \cdot \mathrm{median}(\Delta x_{\mathrm{peaks}})}{W_{\mathrm{core}}} \tag{3}$$

where $W_{\mathrm{core}} = 0.3\,\lambda_J$.

12 Reproduction Instructions

12.1 Step 1: Install Athena++

"'bash git clone https://github.com/PrincetonUniversity/athena-cd athena git checkout v21.1 ./configure --enable-dependencies make -j8 "'

12.2 Step 2: Configure Problem Generator

Copy problem specification from this document into 'athena++/src/prob/filament.cpp'.

12.3 Step 3: Run Simulation

"'bash ./athena++ -i ¡problem$_configfile$ $> -d <$ $output_directory >$ "'

12.4 Step 4: Analyze Output

Use Python analysis scripts provided in supplementary materials or apply peak detection method described in Section ??.

13 Campaign Summary Table

Table 9: Complete campaign summary

Campaign	Sims	f range	β range	Key Result
Field Geometry	314	1.5–3.0	0.3–5.0	$\lambda/W(\theta)$: geometry dominates
DTC	540	1.4–2.2	0.3–1.3	Timescale boundary mapped
Supercritical	654	1.1–3.0	0.3–5.0	Radial collapse dominates
CTZM	96	1.2–1.5	0.3–2.0	Smooth evolution, 100% beading
Perp-Field	96	1.2–1.5	0.3–2.0	Validates CTZM at $f = 1.5$
Turbulence	36	1.0–1.2	0.5–2.0	$<$ 5% λ/W difference
EOS	48	1.0–1.2	0.5–2.0	2.8% variation in λ/W
Resolution	12	1.5–3.0	1.0	7.2% difference, same classification
Total	**2000**			

14 Notes

1. All simulations use periodic boundary conditions on x_1 (longitudinal)
2. Transverse boundary conditions are outflow (zero-gradient)
3. Random seeds control initial perturbation phase
4. Simulation timeout: 600 seconds wall-clock (standard) or extended for specific campaigns
5. Output format: HDF5 with full 3D fields and selected 1D/2D diagnostics
6. Coordinates: x_1 = longitudinal axis, x_2, x_3 = transverse axes

15 References

Athena++ code: Stone et al. (2020), *Athena++ User Guide*, `https://github.com/PrincetonUniversity/athena-`

Filament problem generator: Custom implementation for cylindrical filaments with self-gravity.